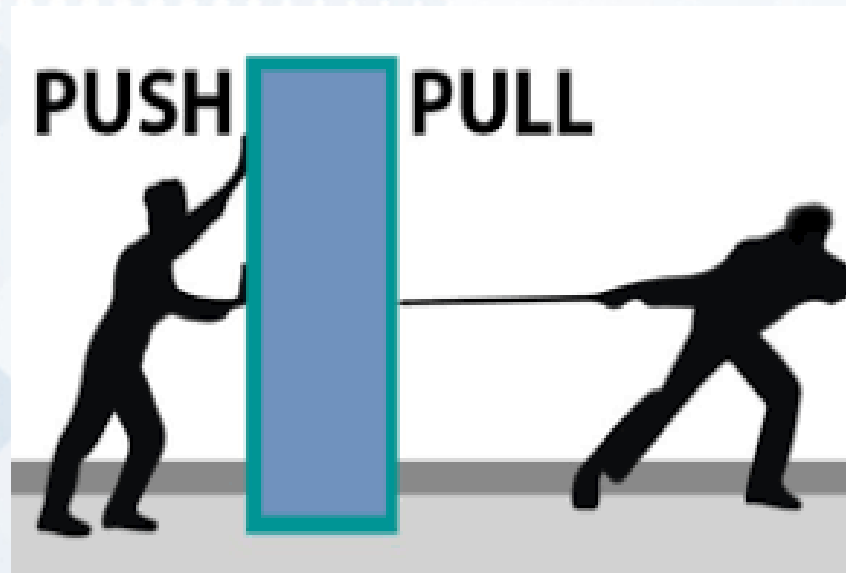


Introduction



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TOPICS COVERED IN PRESENTATION

- ▶ **Basic Concepts**
- ▶ **Force and Characteristics of Force**
- ▶ **System of Force**
- ▶ **Composition of Forces**
- ▶ **Resolution of Forces**
- ▶ **Principle of Transmissibility**
- ▶ **Principle of Superposition**
- ▶ **Law of Gravitation**
- ▶ **Law of Parallelogram**
- ▶ **Newton's Law of Motion**

BASIC CONCEPTS

Space:

- It is a region in all directions encompassing the universe. It is a geometric position occupied by bodies. These positions are described by linear or angular measurements with reference to a defined system of co-ordinates.

Time:

- Time is a measurement to measure a duration between successive events. In the study of statics time does not play important role. In dynamics time is very important parameter. In all system of units, unit of time is second.

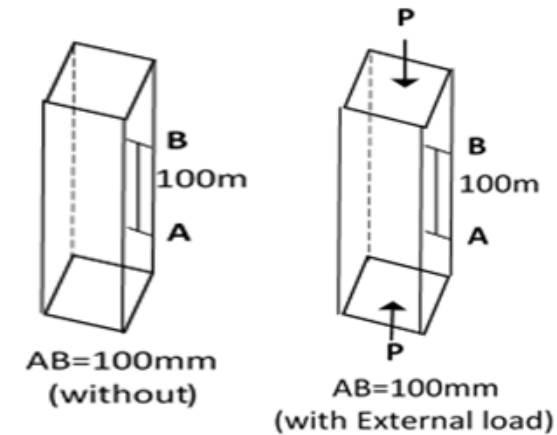
Particle:

- A particle is ideally dimensionless. But it has a very small mass.

BASIC CONCEPTS

Rigid Body:

- A body can be considered as rigid, when the dimensions of body does not change due to application of external force on the body. OR
- In other world, the body which does not deform under applied external force, is called a rigid body.
- Practically no such body exist in nature, but when the change in dimensions are negligible in comparison to the least dimension of body, the body can be idealized as a rigid body.



Deformed Body:

- When a body deforms due to a force or a torque it is said deformable body. Material generates stresses against deformation.

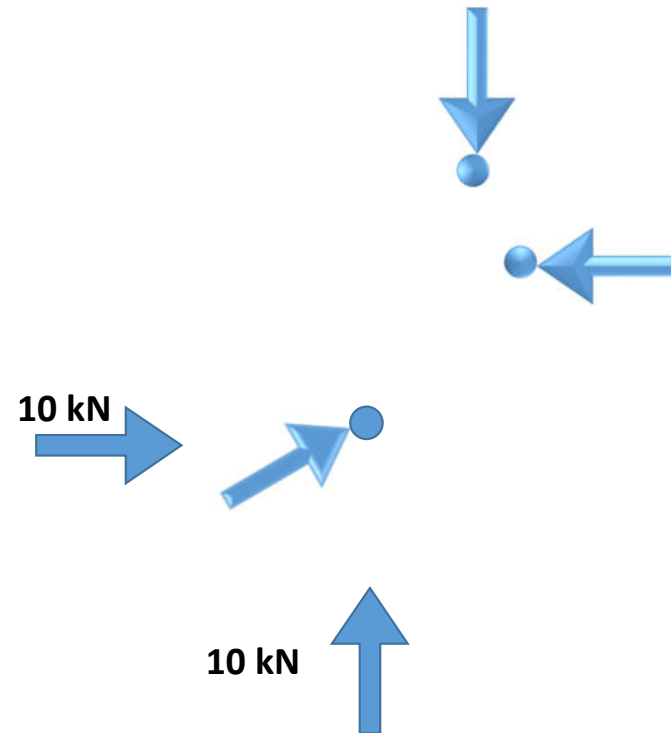
FORCE AND CHARACTERISTICS OF A FORCE

Force:

- Force is an agent, which generates or tends to generate and destroy or tends to destroy the motion in a body

❖ Characteristics of a Force:

- It has a magnitude
- It has a direction
- It has a point of application
- It has a nature;
 - Tensile Force
 - Compressive Force
 - Pull Force
 - Push Force

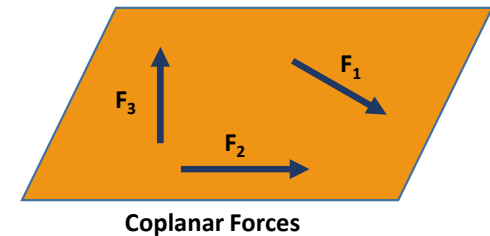


SYSTEM OF FORCES

When two or more forces act on a body, they are called as a system of forces.

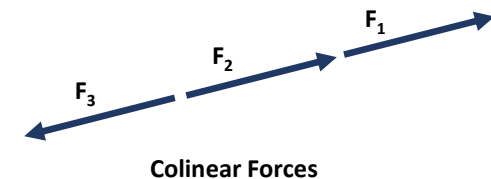
1. Coplanar Forces:

- The forces, whose lines of action lie on the same plane, are known as coplanar forces.



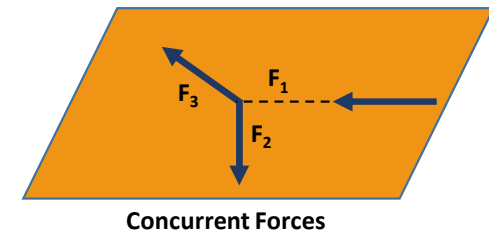
2. Collinear Forces:

- The forces, whose lines of action lie on the same line, are known as collinear forces.



3. Concurrent Forces:

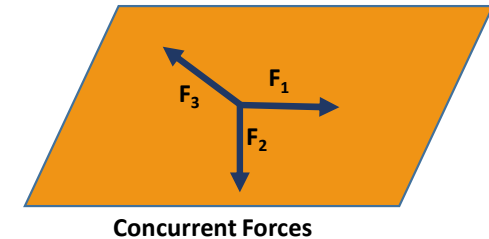
- The forces, whose line of action meet at one point, are known as concurrent forces. The concurrent forces may or may not be collinear.



SYSTEM OF FORCES

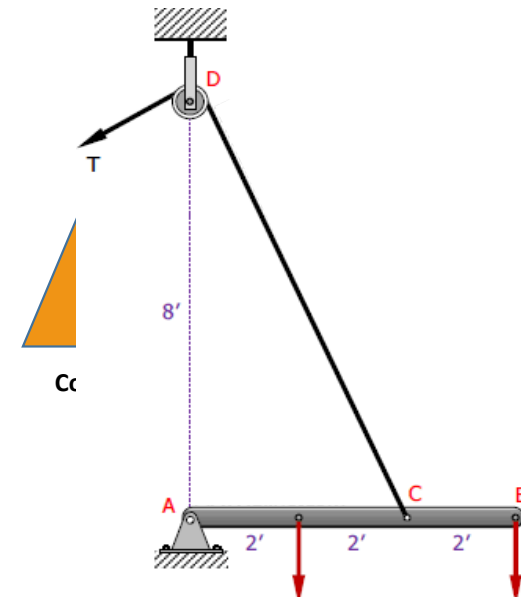
4. Coplanar Concurrent Forces:

- The forces, which meet at one point and their line of action also lie on the same plane, are known as coplanar concurrent forces.



5. Coplanar Non-Concurrent forces:

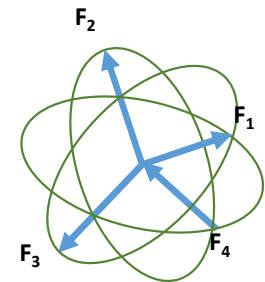
- The forces, which do not meet at one point, but their lines of action lie on the same plane, are known as coplanar non-concurrent forces.



SYSTEM OF FORCES

6. Non-Coplanar Concurrent Forces:

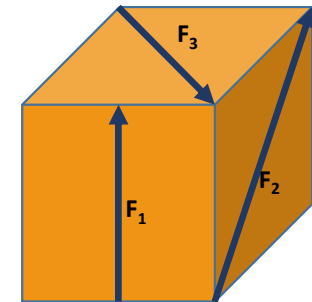
- ➔ The forces, which meet at one point, but their lines of action do not lie on the same plane, are known as non-coplanar concurrent forces.



Non-Coplanar Concurrent Forces

7. Non-Coplanar Non-Concurrent Forces:

- The forces, which do not meet at one point and their lines of action do not lie on the same plane, are called non-coplanar non-concurrent forces.



Non-Coplanar Concurrent Forces

COMPOSITION OF FORCES

When two or more forces are added, is called composition of forces.

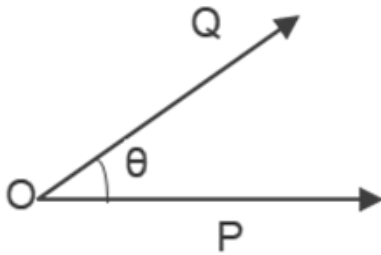


Figure No. 1

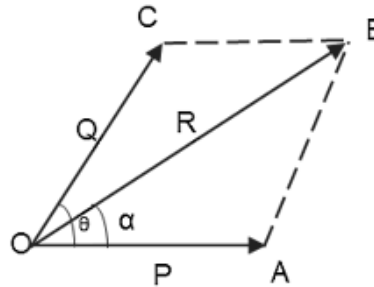


Figure No. 2

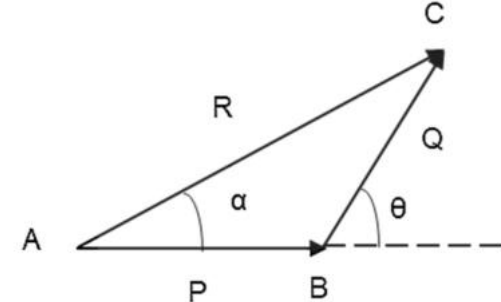


Figure No. 3

- Two force P and Q as shown in figure No. 1, are composed by, the law of parallelogram shown in figure No. 2 and law of triangle shown in Figure No.3, So we get the resultant force R.
- But in case of more than two forces, repetition of these laws done for addition of forces or Polygons law can be used.

RESOLUTION OF FORCES

- To divide the force into components is called resolution of a given force. The force can be resolved into orthogonal components i.e. the components are perpendicular to each other or into oblique components

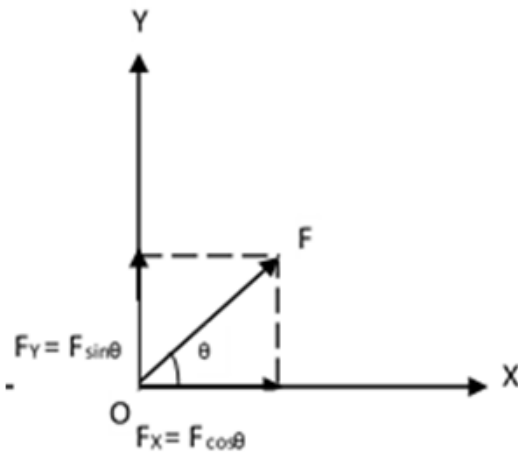


figure 1

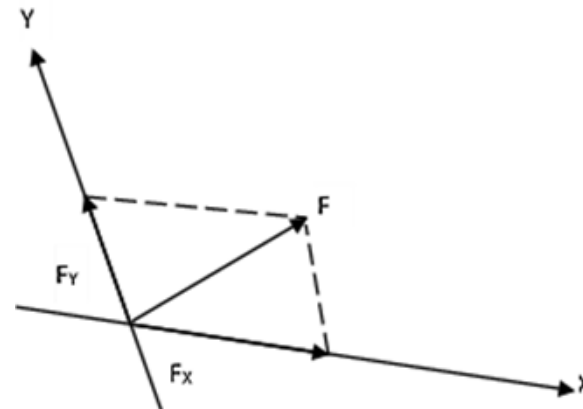
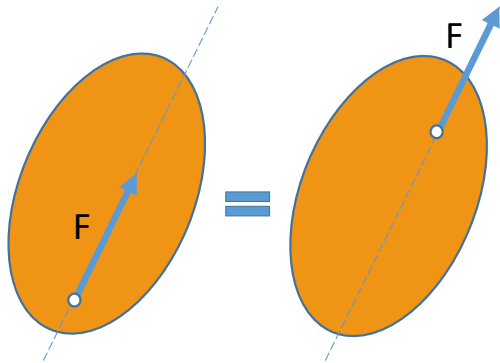


figure 2

- Let the force F is resolved into two orthogonal components as F_X and F_Y as shown in figure 1 and it can be resolved into oblique components as shown in figure 2

PRINCIPLE OF TRANSMISSIBILITY

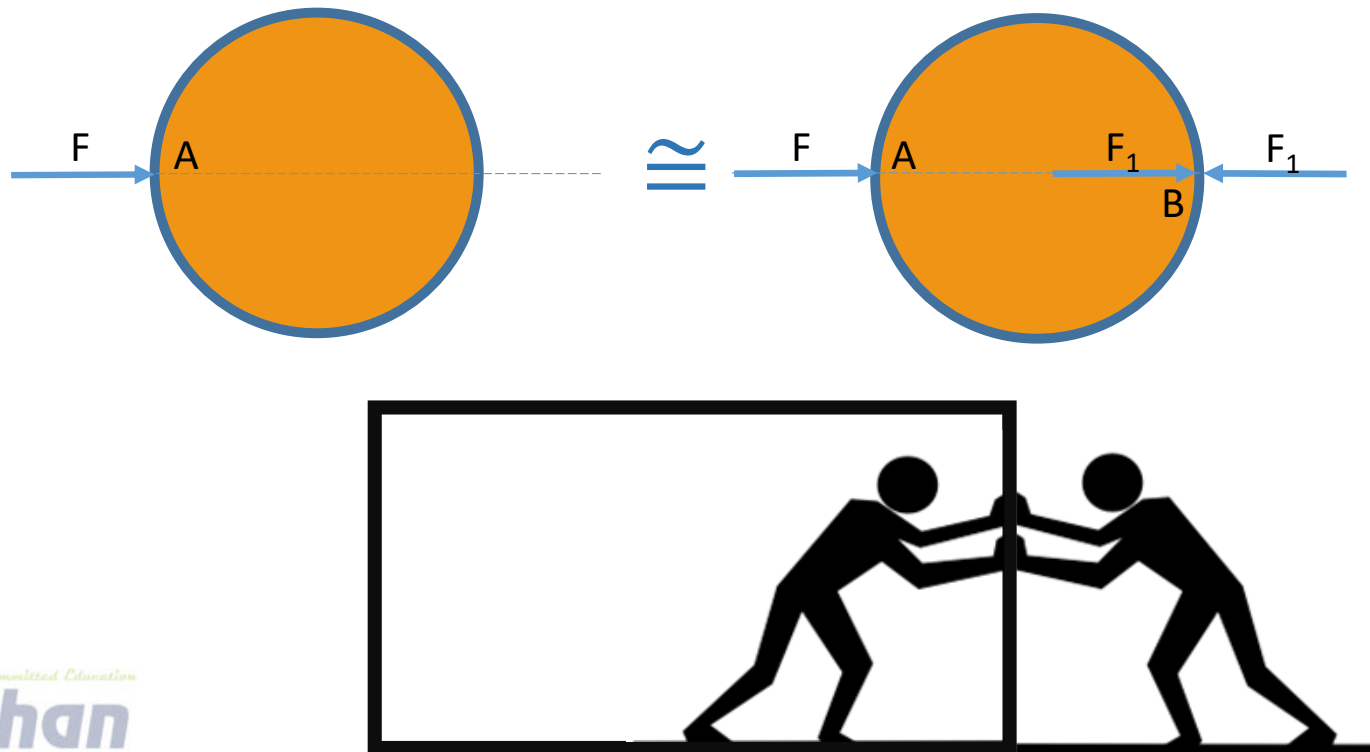
- The principle of transmissibility states that a force may be applied at any point on its given line of action without altering the resultant effects of the force external to the rigid body on which it acts.



- Thus, whenever we are interested in only the resultant external effects of a force, the force may be treated as a sliding vector, and we need specify only the magnitude, direction, and line of action.

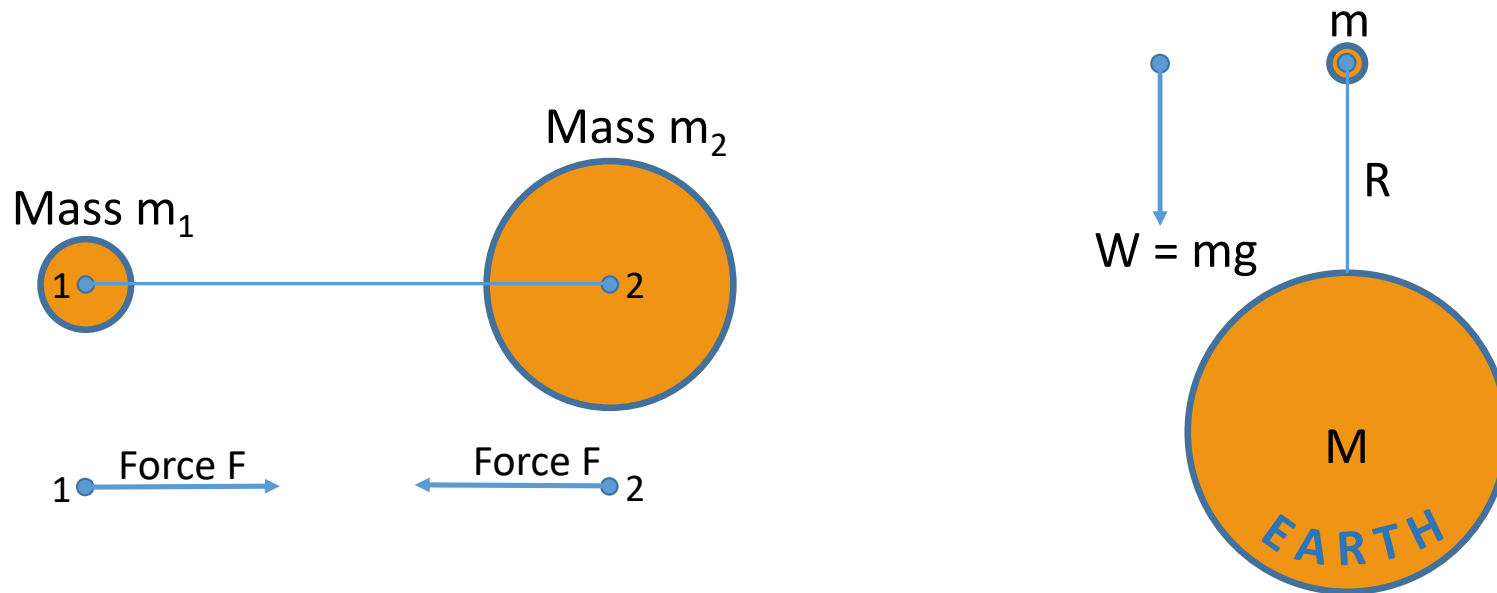
PRINCIPLE OF SUPERPOSITION

- The effect of a force on a body remains same or remains unaltered if a force system, which is in equilibrium, is added to or subtracted from it.



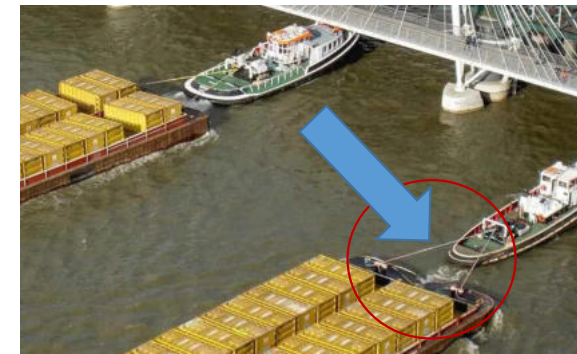
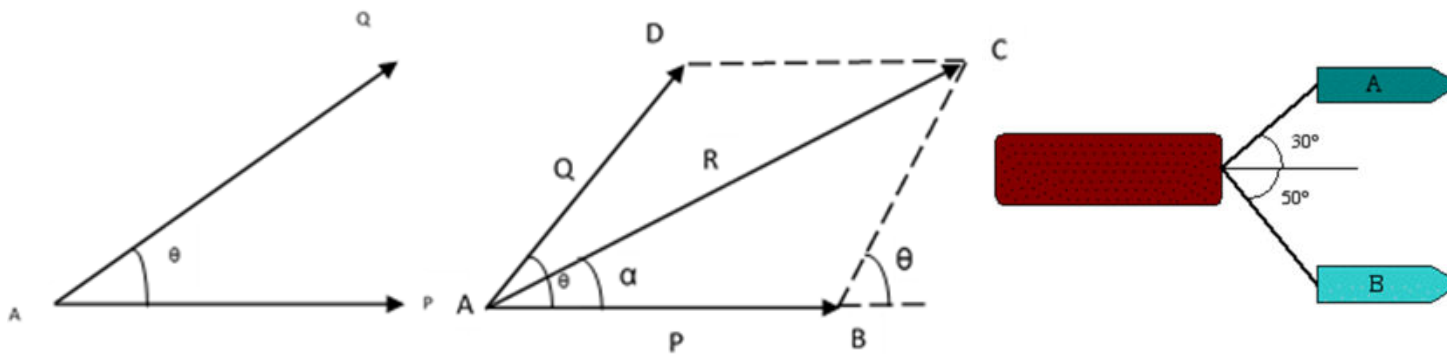
LAW OF GRAVITATION

- Magnitude of gravitational force of attraction between two particles is proportional to the product of their masses and inversely proportional to the square of the distance between their centers.



LAW OF PARALLELOGRAM OF FORCES

- The law of parallelogram states that two forces acting on a particle are taken as two adjacent sides of a parallelogram, then the diagonal passing through the intersection of two forces gives the resultant.



- As shown in figure, P and Q are the two forces acting on a body, are represented as two adjacent sides of a parallelogram ACBD, then the diagonal AC represents the resultant R.

$$R = \sqrt{P^2 + Q^2 + 2PQ\cos\theta}$$

$$\tan\alpha = \frac{Q\sin\theta}{P + Q\cos\theta}$$

NEWTON'S LAWS OF MOTIONS

Newton's First Law

- If the resultant force on a particle is zero, the particle will remain rest(if originally at rest) or will move with constant velocity in a straight line(if originally in motion).

$$\Sigma \mathbf{F} = \mathbf{0}, \quad \text{where } \Sigma \mathbf{F} = \text{resultant force of the system.}$$

Newton's Second Law

- If the resultant force acting on a particle is not zero then the particle will have an acceleration, in the direction of resultant force and it is directly proportional to resultant force.

► Mathematically it is represented as, $\Sigma \mathbf{F} \propto \mathbf{a}$, $\mathbf{F} = m\mathbf{a}$,

Where, $\Sigma \mathbf{F}$ = resultant force, m = mass and a = acceleration.

Newton's Third Law

- The forces of action and reaction between bodies in contact will have same magnitude, same line of action and will be in a direction opposite to each other.